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DEPARTMENT OF STATISTICS

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Obesity: A Statistical Perspective

Exploring Risk Factors Through Logistic Regression

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Supervisor: Professor Sancharee Basak

Declaration:

"I affirm that I have identified all my sources and that no part of my dissertation paper uses unacknowledged materials."

Signature of Student

Department of Statistics

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I would like to express my heartfelt gratitude to my supervisor, Professor Sancharee Basak, for her invaluable support, guidance, and insightful feedback throughout the course of this research. Her expertise and encouragement played a crucial role in shaping my understanding and approach to this study.

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INTRODUCTION

Obesity has become one of the most urgent international public health issues of the 21st century. As per the World Health Organization (WHO), global obesity has almost tripled since 1975, and more than 650 million adults were obese in 2016. The condition is a leading cause of chronic diseases like diabetes, cardiovascular diseases, and some cancers, contributing to higher healthcare expenditure and poorer quality of life. With its multifactorial aetiology—genetic, behavioural, environmental, and socioeconomic—there is a need for an evidence-based approach to determine key determinants and how best to intervene.

This research utilizes sophisticated statistical methods to examine the determinants of obesity, with an emphasis on logistic regression, hypothesis testing, and predictive modelling. The data consist of a range of lifestyle and demographic factors, such as dietary patterns, levels of physical activity, socioeconomic status, and genetic factors. Using statistical tools like odds ratio analysis, Yule's Q, t-tests, and proportion tests, we evaluate the importance of various predictors in affecting obesity. This research aims to identify the most significant lifestyle and demographic predictors of obesity, provide a robust statistical framework for obesity risk assessment, support public health policies with empirical evidence on modifiable risk factors, enhance early intervention strategies through predictive modelling.

By integrating statistical analysis with public health insights, this study seeks to contribute to the global effort in combating obesity and its associated health burdens. The findings will be valuable for healthcare providers, policymakers, and researchers working on preventive health measures.

OBJECTIVE

In the modern era, obesity is one of the most pressing public health challenges, with its effect rising significantly across all age groups. It is a complex condition influenced by multiple factors including dietary habits, physical activity levels, genetic predisposition, and socio-economic conditions. Obesity is known to increase the risk of severe health complications such as cardiovascular diseases, diabetes, hypertension, and metabolic disorders, making it a major concern for global healthcare systems.

By analysing obesity through a data centric approach, this dissertation aims to provide valuable approaches that can contribute to preventive healthcare strategies. The findings can be useful for policymakers, healthcare professionals, and individuals seeking to understand and mitigate obesity-related risks.

DATA SOURCE

The OBESITY PREDICTION dataset is collected from the website Kaggle.com. The link to the original data is attached below.

https://www.kaggle.com/datasets/ruchikakumbhar/obesity-prediction

DATA DESCRIPTION

The OBESITY PREDICTION dataset include data for the estimation of obesity levels in individuals from the countries of Mexico, Peru, and Colombia, based on their physical conditions and daily habits.

The data contains 17 columns and 2111 rows. The 17 columns can work as the response and predictors in our analysis. The variables are described below.

1. **Gender:** Gender
2. **Age**: Age
3. **Height** : in metres
4. **Weight** : in kgs
5. **family_history** : Has a family member suffered or suffers from overweight?
6. **FAVC** : Do you eat high caloric food frequently?
7. **FCVC** : Do you usually eat vegetables in your meals?
8. **NCP** : How many main meals do you have daily?
9. **CAEC** : Do you eat any food between meals?
10. **SMOKE** : Do you smoke?
11. **CH2O** : How much water do you drink daily?
12. **SCC** : Do you monitor the calories you eat daily?
13. **FAF**: How often do you have physical activity?
14. **TUE** : How much time do you use technological devices such as cell phone, videogames, television, computer and others?
15. **CALC** : How often do you drink alcohol?
16. **MTRANS** : Which transportation do you usually use?
17. **Obesity_level (Target Column)** : Obesity level

Transformation of variable for our research purpose:

For this dissertation, I have chosen 10 out of these 17 variables and renamed them according to my convenience. The new nomenclature of the variables and their description is written below.

1. **obesity**: A binary response indicating the person is obese or not.
2. **gender**: A binary variable indicating the person is male or female.
3. **age**: Age of the person.
4. **height**: Height (in metre).
5. **weight**: Weight (in kg).
6. **family_history**: Binary variable indicating the presence of obesity among the past family members, i.e. obesity was present or not in family.
7. **high_calorie**: Binary variable indicating the diet of the person, i.e. high calorie, or low calorie.
8. **smoke**: Binary variable indicating the smoking status of the person.
9. **physical_activity**: Numeric variable describing the number of times the person gets involved in physical activity per day.
10. **alcohol**: Ordinal categorical variable denoting the extent of alcohol consumption of the person. Here, never, sometimes, and frequent.

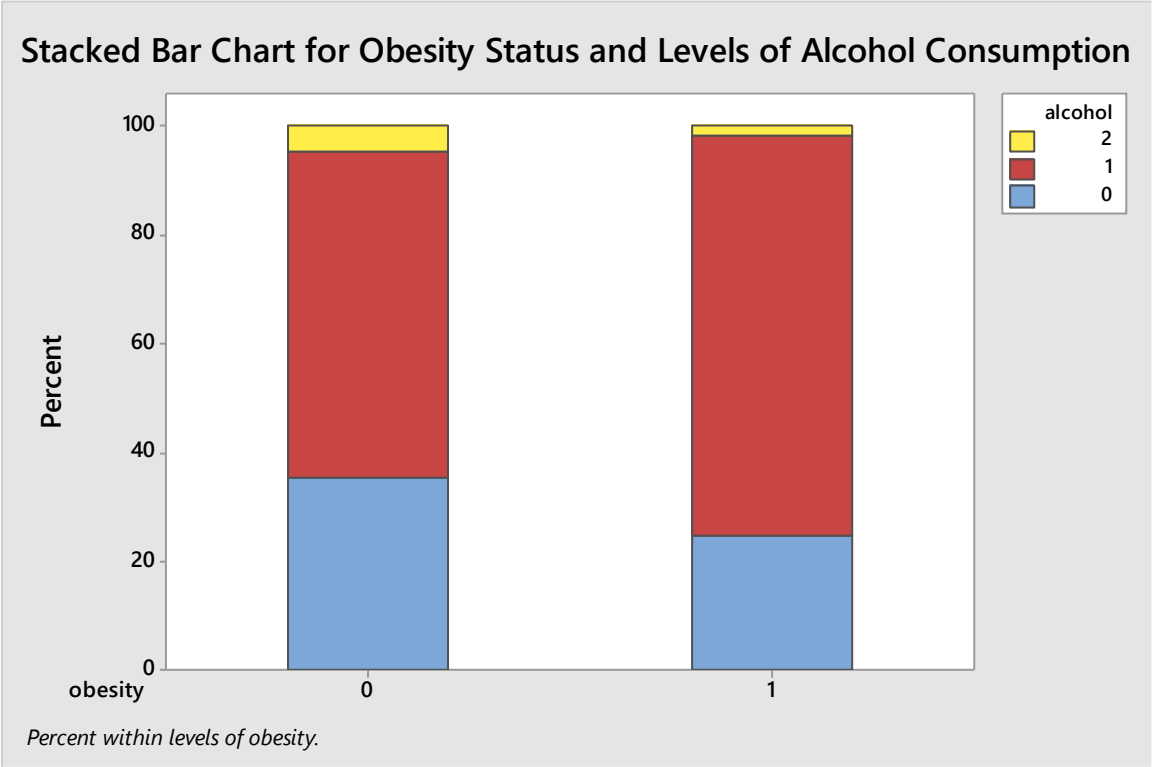
Here we have **obesity** as response variable which takes values True (1) and False (0). And the rest of the nine variables are used to predict if a person is obese or not.

After removing blank elements corresponding to any row or column and cleaning the data we are left with 2103 rows. Each row represents the data collected for a specific human being. Out of the 2103 people, **1138 are not obese and 965 are obese**.

DATA VISUALISATION

To explore the relationship between obesity status and different other variables, we used graphical methods such as bar charts. These visualizations help in understanding the distribution of obesity across different predictors. These plots provide an initial insight into potential associations before conducting statistical tests.

1. **Stacked Bar Chart for Obesity Status and Levels of Alcohol Consumption**



This chart illustrates the relationship between obesity levels and alcohol consumption levels, expressed as percentages. The x-axis represents obesity levels (0 for non-obese and 1 for obese), while the y-axis shows the percentage

of the population. The chart is divided into three coloured segments, each corresponding to a level of alcohol consumption (0 to 2),

Where 0 means no alcohol consumption, 1 means sometimes and 2 means very frequent consumption.

The detailed interpretation:

A. Obesity level 0:

About 35% of non-obese persons do not intake alcohol (blue). Around 60% of non-obese persons consume alcohol sometimes (red). Then only 5% of the non-obese population intakes alcohol frequently (yellow).

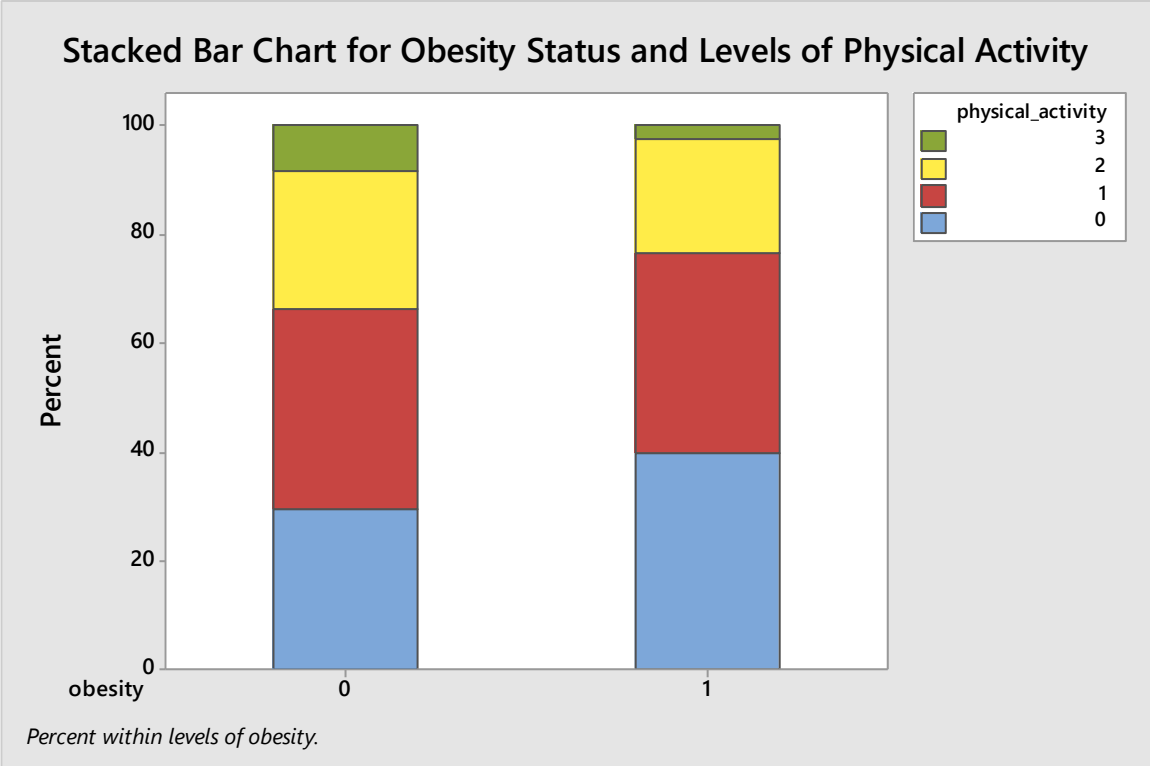
B. Obesity level 1:

About 25% of obese persons do not intake alcohol (blue). Around 72% of obese persons consume alcohol sometimes (red). Then only 3% of the obese population intakes alcohol frequently (yellow).

Comment:

The data suggests that **obese individuals are slightly more likely to consume higher levels of alcohol than non-obese individuals**. However, the relationship is not extreme. This insight could inform public health strategies targeting both alcohol use and obesity prevention.

2. Stacked Bar Chart for Obesity Status and Levels of Physical Activity



This chart illustrates the relationship between obesity levels and physical activity levels within a population, expressed as percentages. The x-axis represents obesity levels (0 for non-obese and 1 for obese), while the y-axis shows the percentage of the population. The chart is divided into four coloured segments, each corresponding to a level of physical activity (0 to 3), as indicated by the legend on the right. Where each number denotes the number times individuals are engaged in physical activity. The detailed interpretation:

A. Obesity level 0:

About 30% of non-obese persons engage in no physical activity (blue). Around 40% of non-obese persons engage in physical activity once in a day (red). Then 20% of the non-obese population performs physical activity twice a day (yellow). And in the end about 10% of non-obese persons are highly physically active.

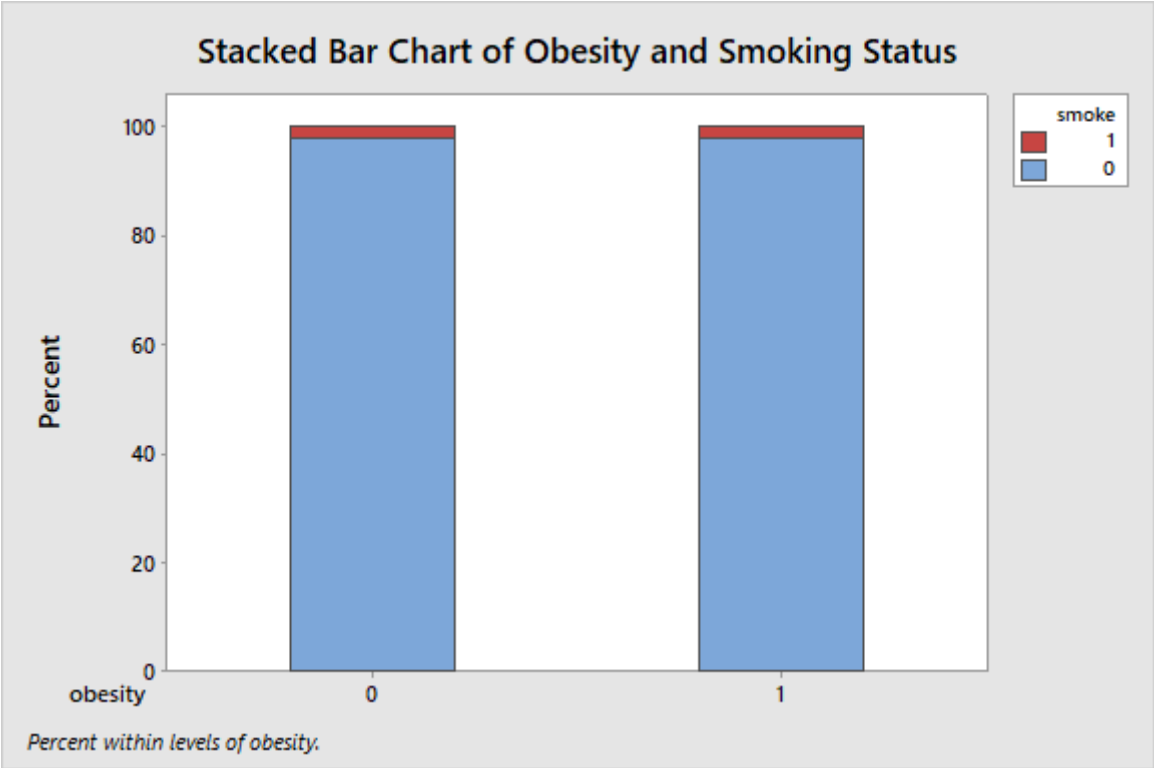
B. Obesity level 1:

About 50% of obese persons engage in no physical activity (blue). Around 30% of obese persons engage in physical activity once in a day (red). Then 15% of the obese population performs physical activity twice a day (yellow). And in the end only 5% of obese persons are highly physically active.

Comment:

The chart suggests that **obesity is associated with lower levels of physical activity**. Obese individuals seem to be more inactive and is not likely to perform high level physical activities compared to that of non-obese individuals. This imply that increase in physical activity might be a key factor in assessing obesity.

3. Stacked Bar Chart for Obesity Status and Smoking Status



This chart illustrates the relationship between obesity levels and smoking status, expressed as percentages. The x-axis represents obesity levels (0 for non-obese

and 1 for obese), while the y-axis shows the percentage of the population. The chart is divided into two coloured segments, each corresponding to a smoking status. Where 0 means non-smokers, 1 means regular smokers.

The detailed interpretation:

A. Obesity level 0:

About 95% of non-obese persons do not smoke (blue) and only 5% of non-obese persons smoke (red).

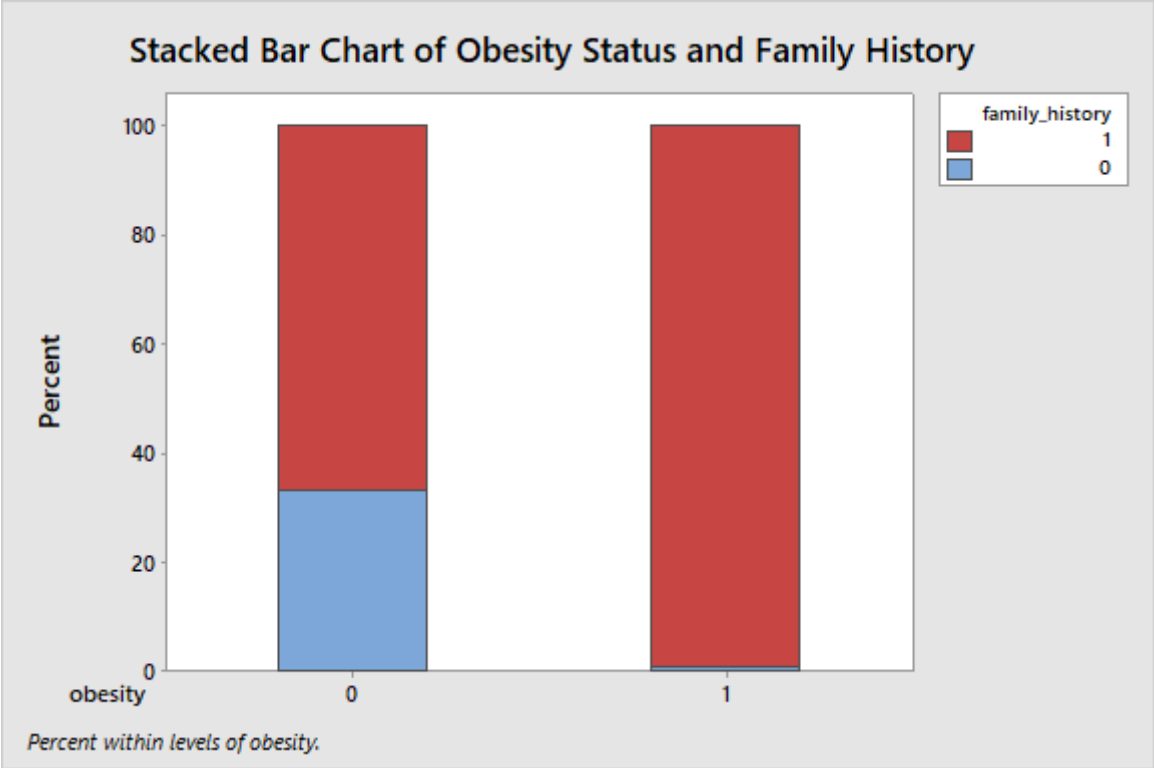
B. Obesity level 1:

About 95% of obese persons do not smoke (blue) and only 5% of obese persons smoke (red).

<u>Comment:</u>

The data suggests that there is **no noticeable association** between smoking status and obesity status.

4. Stacked Bar Chart for Obesity Status and Family History



This chart illustrates the relationship between obesity levels and family history, expressed as percentages. The x-axis represents obesity levels (0 for non-obese and 1 for obese), while the y-axis shows the percentage of the population. The chart is divided into two coloured segments, each corresponds to family history. Where 0 means absence of obesity in past family members, 1 means presence of obesity in past family members.

The detailed interpretation:

A. Obesity level 0:

About 30% of non-obese persons do not have obesity in their family history. (blue) and about 70% of non-obese persons have a family history of obese members. (red).

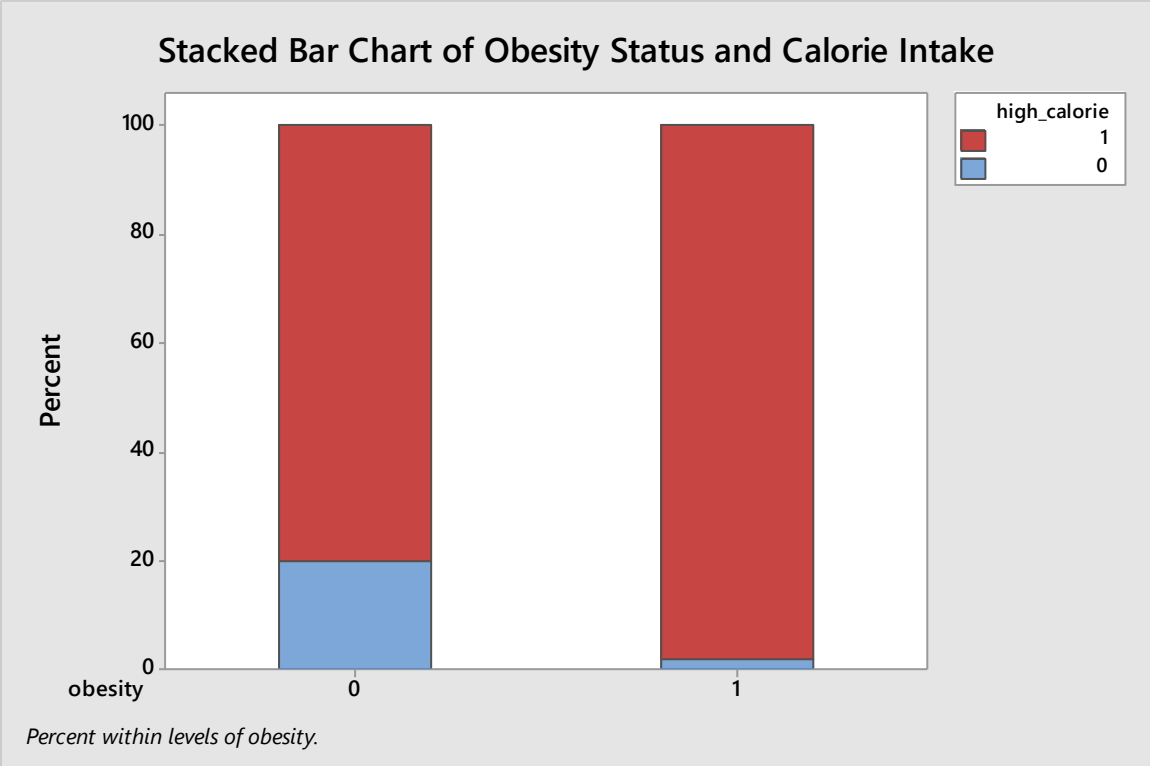
B. Obesity level 1:

About 3% of obese persons do not have obesity in their family history. (blue) where about 97% of non-obese persons have a family history of obese members. (red).

<u>Comment:</u>

There is a **strong association** between being obese and having a family history of obesity. The presence of a family history of obesity is a strong indicator of being obese in this dataset. In summary, this stacked bar chart demonstrates a clear association between obesity and family history. Individuals with a family history of obesity are significantly more likely to be obese.

5. Stacked Bar Chart for Obesity Status and Calorie Intake



This chart illustrates the relationship between obesity levels and family history, expressed as percentages. The x-axis represents obesity levels (0 for non-obese and 1 for obese), while the y-axis shows the percentage of the population.

The chart is divided into two coloured segments, each corresponds to calorie intake.

Where 0 means low calorie diet, 1 means high calorie diet.

The detailed interpretation:

A. Obesity level 0:

About 20% of non-obese persons intake low calorie diet (blue) and about 70% of non-obese persons intake high calorie diet. (red).

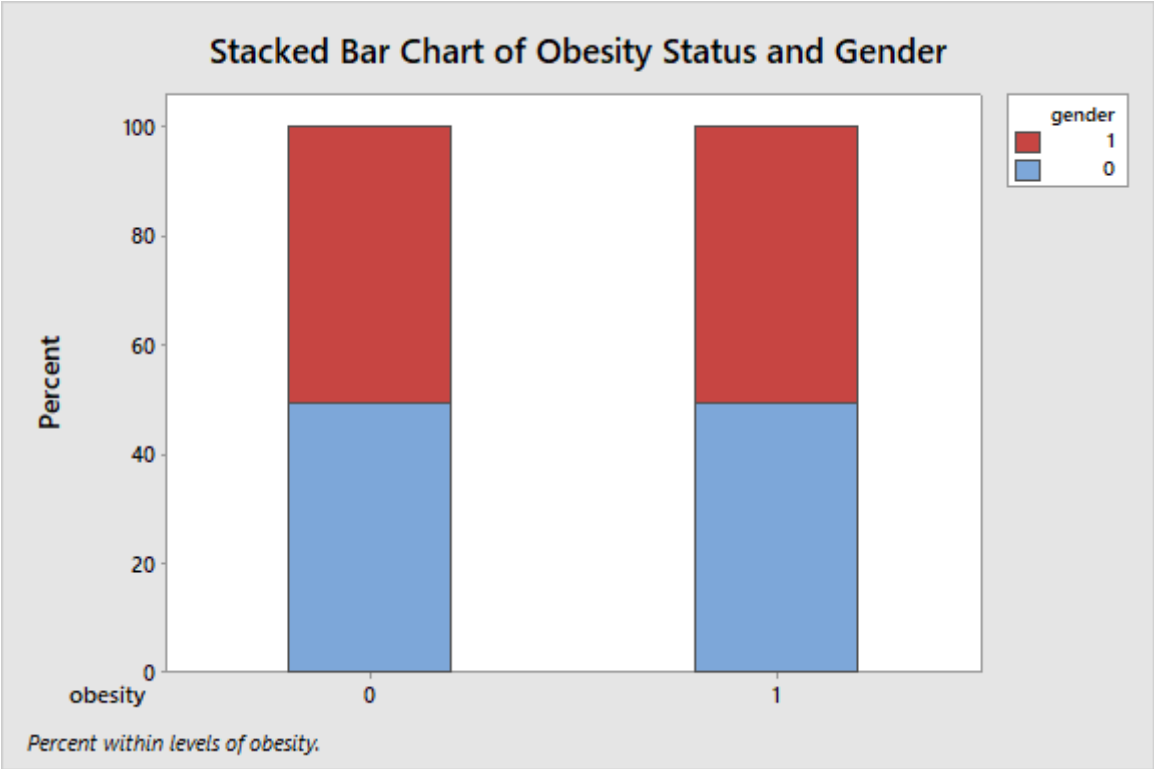
C. Obesity level 1:

About 4% of obese persons intake low calorie diet (blue) and about 96% of obese persons intake high calorie diet. (red).

Comment:

There is a **strong association** between being obese and having a high calorie diet. Intake of high calorie is a strong indicator of being obese in this dataset. In summary, this stacked bar chart demonstrates a clear association between obesity and calorie intake. Individuals with high calorie diet are significantly more likely to be obese.

6. Stacked Bar Chart of Obesity Status and Gender



This chart illustrates the relationship between obesity levels and gender, expressed as percentages. The x-axis represents obesity levels (0 for non-obese and 1 for obese), while the y-axis shows the percentage of the population. The chart is divided into two coloured segments, each corresponding to a smoking status. Where 0 means female, 1 means male.

The detailed interpretation:

A. Obesity level 0:

About 50% of non-obese persons are female (blue) and 50% of non-obese persons are male (red).

B. Obesity level 1:

About 50% of obese persons are female (blue) and 50% of obese persons are male (red).

<u>Comment:</u>

The data suggests that there is **no noticeable association** between gender and obesity status. Gender does not appear to be a significant factor in distinguishing between non-obese and obese individuals, as both categories have approximately equal proportions of males and females.

MEASURES OF ASSOCIATION

To measure the association between the response **obesity** and other categorical predicting variables, we form separate contingency tables. Each table represents the different categories of the respective variables in such a way that it becomes easier to interpret and compare between the association of the predictors with the response.

We take **Yule's Measure of Association** which is denoted by **Q** as an appropriate measure of association for 2x2 tables, which is given by,

$$Q=(ad-bc)/(ad+bc)$$

a: element corresponding to the 1st row and 1st column

b: element corresponding to the 1st row and 2nd column

c: element corresponding to the 2nd row and 1st column

d: element corresponding to the 2nd row and 2nd column

1. **An Appropriate 2x2 Contingency Table for Obesity Status and Gender**

Gender Obesity	Female	Male	Total
Non Obese	**562**	**576**	**1138**
Obese	**476**	**489**	**965**
Total	**1038**	**1065**	**2103**

The columns represent the Gender and the rows represent the obesity status.

Now for this table, Yule's measure is given by,

$Q_1 = 0$ (approximately)

Comment:

There is no association between the two binary variables obesity status and gender. In other words, the presence or absence of one variable does not provide any predictive power for the presence or absence of the other. The two variables are **statistically independent.**

2. **An Appropriate 2x2 Contingency Table for Obesity Status and Family History**

Family History Obesity	No	Yes	Total
Non Obese	**377**	**761**	**1138**
Obese	**8**	**957**	**965**
Total	**385**	**1718**	**2103**

The columns represent the Family History and the rows represent the obesity status.

Now for this table, Yule's measure is given by,

$Q_2 = 0.97$ (approximately)

Comment:

There is a **strong positive association** between Obesity status and Family history. This suggests a high level of dependence between the two variables, implying that knowing the state of one variable gives a strong indication of the state of the other.

3. **An Appropriate 2x2 Contingency Table for Obesity Status and Dietary Habit**

Diet Obesity	Low Calorie	High Calorie	Total
Non Obese	**226**	**912**	**1138**
Obese	**19**	**946**	**965**
Total	**245**	**1858**	**2103**

The columns represent the dietary habit and the rows represent the obesity status.

Now for this table, Yule's measure is given by,

$Q_3 = 0.85$ (approximately)

Comment:

There is a **strong positive association** between obesity status and dietary habit. This suggests a high level of dependence between the two variables, implying that knowing the state of one variable gives a strong indication of the state of the other.

4. **An Appropriate 2x2 Contingency Table for Obesity Status and Smoking Status**

Smoking Obesity	No	Yes	Total
Non Obese	**1116**	**22**	**1138**
Obese	**943**	**22**	**965**
Total	**2059**	**44**	**2103**

The columns represent the Smoking Status and the rows represent the obesity status.

Now for this table, Yule's measure is given by,

$Q_4 = 0.08$ (approximately)

<u>Comment:</u>

There seems to be no association between obesity status and smoking status. In other words, the presence or absence of one variable does not provide any predictive power for the presence or absence of the other. The two variables are **statistically independent.**

- Now for 2x3 contingency tables (here **alcohol** and **physical_activity** have got 3 levels and 4 levels respectively), we cannot use Yule's measure of association. So, we use the Chi square test for independence to check whether the two variables are related to obesity status or not.

Procedure of the Chi Square Test:

Step 1: Organize the data into a contingency table. And proceed to test the null hypothesis: independence of the two variables against the alternative hypothesis which suggests dependency.

Step 2: Calculate the Chi-Square statistic using the formula:

$$\chi^2 = \sum_{i=1}^{p}\sum_{j=1}^{q}\{f_{ij} - (f_{i0}\, f_{0j})/n\}^2/(\frac{f_{i0}f_{0j}}{n})$$

Where f_{ij} be the number of members belonging to the ith row and jth column of the table. f_{i0} is the ith row sum and f_{oj} is the jth column sum. i=1(1)p and j=1(1)q.

Step 3: Determine the degrees of freedom: df=(p−1)(q−1), where (p) is the number of rows (2 for obesity) and (q) is the number of columns (levels of the categorical variable).

Step 4: Compare the test statistic to a Chi-Square distribution (using a p-value or critical value) to assess significance.

5. **An Appropriate 2x3 Contingency Table for Obesity Status and Levels of Alcohol Consumption**

Alcohol Obesity	Never	Sometimes	Frequently	Total
Non Obese	**402**	**682**	**54**	**1138**
Obese	**237**	**712**	**16**	**965**
Total	**639**	**1394**	**70**	**2103**

The columns represent the Alcohol consumption levels and the rows represent the obesity status.

Now the chi square statistic for the above table is given by,

$\chi^2 = 49.987$

The degrees of freedom of this test statistic = 2

The critical value for a 5% level of significance is 5.991.

Since the observed chi square statistic is greater than the critical value, **we reject the null.**

<u>Comment</u>:

In light of the given data, obesity status and levels of alcohol consumption seems to be **associated** with each other which is **statistically significant.**

6. **An Appropriate 2x3 Contingency Table for Obesity Status and Levels of Physical Activity**

Physical Activity Obesity	Never	Once	Twice	Thrice	Total
Non Obese	**333**	**419**	**292**	**94**	**1138**
Obese	**385**	**355**	**200**	**25**	**965**
Total	**718**	**774**	**492**	**119**	**2103**

The columns represent the Physical Activity levels and the rows represent the obesity status.

Now the chi square statistic for the above table is given by,

$\chi^2 = 52.393$

The degrees of freedom of this test statistic = 3

The critical value for a 5% level of significance is 7.815.

Since the observed chi square statistic is greater than the critical value, **we reject the null.**

<u>Comment</u>:

In light of the given data, obesity status and levels of physical activity seems to be **associated** with each other which is **statistically significant.**

PARITY BETWEEN DATA VISUALISATION AND MEASURES OF ASSOCIATION

The findings from both association measures and data visualization are all consistent. This trend is observed for all the variables under consideration, further ensuring the validity of statistical associations made in contingency tables by the findings based on visual representation. So, the parity holds and we can conclude the data in hand is consistent.

SOME TESTING PROCEDURES

We perform some testing procedures for the continuous variables' height, weight, and age.

1. **Bootstrap Symmetry Test:**

   We apply the symmetry.test( ) function from the "lawstat" package in R. We perform the test twice. Once for weight and once for height. The test checks whether the variable is symmetric about its median or not.

   The null hypothesis: The distribution is symmetric.

```
        m-out-of-n bootstrap symmetry test by Miao, Gel, and Gastwirth (2006)

data:  weight
Test statistic = 7.7892, p-value < 2.2e-16
alternative hypothesis: the distribution is asymmetric.
sample estimates:
bootstrap optimal m
                404
```

Fig: Bootstrap test for Weight

```
        m-out-of-n bootstrap symmetry test by Miao, Gel, and Gastwirth (2006)

data:  height
Test statistic = 0.72406, p-value = 0.394
alternative hypothesis: the distribution is asymmetric.
sample estimates:
bootstrap optimal m
               1477
```

Fig: Bootstrap test for Height

From both the tests we see that, the p value in each case is larger than the level of significance which is 0.05. So, we accept the null hypothesis and conclude that the distribution of height and weight are both symmetric. So we can proceed towards regression including them.

2. Proportionality Test for Age Categories with Respect to Obesity Status:

For this approach, we divide the variable Age in two categories – below 30 and above 30. Now we are **to test**:

$$H_0 : p_1 = p_2 \text{ against } H_1 : \text{not } H_0$$

Where p_1 = the proportion of obese individuals in the **30 and above** group and p_2 = the proportion of obese individuals in the **below 30** group

Test Statistic:

$$Z = \frac{(\widehat{p_1} - \widehat{p_2})}{\sqrt{\widehat{p_1}(1 - \widehat{p_2})(\frac{1}{n_1} + \frac{1}{n_2})}}$$

And for large enough sample sizes, Z under $H_0 \sim N(0,1)$ distribution.

Where, $\widehat{p_1}$ = sample proportion for the first group

$\widehat{p_2}$ = sample proportion for the second group

$\hat{p}$ = pooled proportion

And n_1, n_2 are the sample sizes of the first and the second group respectively.

Test Rule: Reject H_0 iff $|Z_{obs}| > \tau(\alpha/2)$ at α level of significance. Where $\tau(\alpha/2)$ is the upper α/2th point of a N(0,1) distribution.

Computation:

$\widehat{p_1} = 0.5596$, $\widehat{p_2} = 0.4362$, $\hat{p} = 0.4588$, $n_1 = 386$, $n_2 = 1717$

$Z_{obs} = 4.3948$, $\alpha = 0.05$, $\tau_{0.025} = 1.96$

Decision: Since, $Z_{ons} > \tau_{0.025}$, we reject H_0 at 5% level of significance.

Conclusion: Thus, the proportion of obesity differs significantly among the two age groups.

LOGISTIC REGRESSION

Let us denote the response variable **obesity** by Y. Here Y is a binary variable denoting the obesity status of a person.

Y = 1, if a person is obese

0, if a person is not obese

We assume that, $Y_i \sim$ Bernoulli (p_i), i = 1(1)n

Where, p_i denotes the probability of the ith person being obese. i= 1(1)n and n denote the total number of rows in the cleaned dataset. Here, n=2103

Now the rest of the 9 variables X_1, X_2, X_3, X_4, X_5, X_6, X_7, X_8, X_9 be the predictors of our logistic regression model. Where,

X_1: **gender**

X_2 : **age**

X_3: **alcohol (X_{31} and X_{32} denotes the level 1 and 2 of alcohol consumption)**

X_4: **physical_activity (X_{41}, X_{42} and X_{43} denotes the level 1, 2 and 3 of physical activity)**

X_5: **family_history**

X_6: **high_calorie**

X_7: **smoke**

X_8: **weight**

X_9: **height**

The logistic regression is the linear combination of predictors. And it models the log odds of p with respect to different predictors.

Model: y = log $(\frac{p}{1-p})$ = $\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{31} x_{31} + \beta_{32} x_{32} + \ldots\ldots + \beta_9 x_9 + \varepsilon$

In general, y_i = log $(\frac{p_i}{1-p_i})$ = $\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_{31} x_{31i} + \ldots\ldots + \beta_9 x_{9i} + \varepsilon_i$,

i = 1(1)n

In this model, β_0 is the intercept term.

And, β_1, β_2, β_{31}, β_{32} ……, β_9 are the coefficients for the predictors X_1, X_2, X_{31}, X_{32}, X_{41}, X_{42}, X_{43}, X_5, X_6, X_7, X_8, X_9. And all of them are unknown model parameters.

And, ε is the random error associated to the model.

Here log $(\frac{p_i}{1-p_i})$ denotes the log odds of $Y_i = 1$, i = 1(1)n

The probability p_i can be expressed as,

$$p_i = \frac{exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_{31} x_{31i} + \ldots\ldots + \beta_9 x_{9i})}{1 + exp(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_{31} x_{31i} + \ldots\ldots + \beta_9 x_{9i})}, \text{ i} = 1(1)\text{n}$$

Parameter Estimation:

The parameters β_1, β_2, β_3, ……, β_9 are estimated using **Method of Maximum Likelihood.**

The likelihood function for n observations is:

$$L(\beta) = \prod_{i=1}^{n} p_i^{Y_i} (1 - p_i)^{(1-Y_i)}.$$

The log-likelihood function is then given by –

$$\log(L) = \sum_{i=1}^{n} [Y_i * \ln p_i + (1 - Y_i) \ \ln(1 - p_i)]$$

The Score equations are given by,

$$\sum_{i=1}^{n} (Y_i - p_i) = 0.$$

$$\sum_{i=1}^{n} (Y_i - p_i) \ x_{1i} = 0.$$

…..

$$\sum_{i=1}^{n} (Y_i - p_i) \ x_{9i} = 0.$$

We solve the score equations simultaneously and obtain the estimates $\widehat{\beta_0}$, $\widehat{\beta_1}$, $\widehat{\beta_2}$, …. $\widehat{\beta_9}$.

Then we fit a logistic regression on the data and observe the presence of **Multicollinearity.**

Multicollinearity happens when two or more predictor variables within a regression model are correlated highly, which implies that they hold redundant or overlapping information. Multicollinearity makes it challenging to decide the effect of each predictor individually on the response variable.

We use the vif( ) function from the "car" package in R to obtain the **Variance Inflation Factor (VIF)**. In general, VIF > 5 means multicollinearity exists.

The Variance Inflation Factor (VIF) indicates to what extent the variance of a regression coefficient is inflated because of multicollinearity. That is, it informs us if a predictor is highly correlated with other predictors in the model.

We find the VIF values as,

	VIF
gender	3.962980
age	5.136187
height	12.040828
weight	10.368177
family_history	1.093860
high_calorie	5.081522
smoke	1.010983
physical_activity	1.761019
alcohol	2.178954

Since height has a very high VIF value, we remove it from the regression model and fit the model again without the variable X_9.

After removal of height the VIF values of the new model is as follows.

	VIF
weight	1.843896
smoke	1.043000
alcohol	1.034891
gender	1.807106
family_history	1.028449
high_calorie	1.044289
physical_activity	1.029317
age	1.042081

From the new model we can see all the VIF values are small. Hence, we can say **we have successfully omitted the problem of multicollinearity.**

The fitted new logistic regression model is shown below.

```
> summary(model.new)

Call:
glm(formula = obesity ~ weight + smoke + alcohol + gender + family_history +
    high_calorie + physical_activity + age, family = binomial,
    data = data)

Coefficients:
                    Estimate Std. Error z value Pr(>|z|)
(Intercept)        -29.57472    2.12470 -13.919  < 2e-16 ***
weight               0.35766    0.02521  14.185  < 2e-16 ***
smoke1               0.80266    0.88794   0.904 0.366017
alcohol1            -0.61811    0.23538  -2.626 0.008640 **
alcohol2            -1.35660    0.47971  -2.828 0.004685 **
gender              -4.66319    0.39443 -11.823  < 2e-16 ***
family_history1      1.05806    0.58661   1.804 0.071282 .
high_calorie1        0.31963    0.51062   0.626 0.531341
physical_activity1  -1.00636    0.26780  -3.758 0.000171 ***
physical_activity2  -0.42063    0.30226  -1.392 0.164038
physical_activity3  -1.11007    0.44339  -2.504 0.012294 *
age                  0.02193    0.01422   1.542 0.122960
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2901.13  on 2102  degrees of freedom
Residual deviance:  548.29  on 2091  degrees of freedom
AIC: 572.29

Number of Fisher Scoring iterations: 9
```

So, by Fisher Scoring, we get the estimates of the model parameters.

β0	β1	β2	β31	β32	β41	β42	β43
-29.57	-4.66	0.02	-0.62	-1.36	-1.01	-0.42	-1.11

β5	β6	β7	β8
1.06	-0.32	0.80	0.36

So, the fitted model is given by,

$$\log\left(\frac{p}{1-p}\right) = -29.57 - 4.66x_1 + 0.02x_2 - 0.62x_{31} - 1.36x_{32} - 1.01x_{41} - 0.42x_{42} - 1.11x_{43} + 1.06x_5 - 0.32x_6 + 0.8x_7 + 0.36x_8$$

<u>Interpretation of the Model Parameters:</u>

1. $\widehat{\beta_0}$= -29.57 means when all the predictors are zero, the log odds of a person being obese is -29.57 times the log odds of a person being non obese.
2. $\widehat{\beta_1}$ = -4.66 means keeping other factors fixed, if gender changes from male to female, the log odds of obesity decrease by 4.66.
3. $\widehat{\beta_2}$ = 0.02 means keeping other factors fixed, a one-year increase in age increases the log odds of being obese by 0.02.
4. $\widehat{\beta_{31}}$ = -0.62 means keeping other factors constant individuals consuming alcohol sometimes have lower log odds of obesity than non-drinkers.
5. $\widehat{\beta_{32}}$ = -1.36 means keeping other factors constant, individuals consuming alcohol frequently have even lower log odds of obesity than those who drink sometimes.
6. $\widehat{\beta_{41}}$ = -1.01 means keeping other factors constant, engaging in low physical activity decreases log odds of obesity by 1.01.

7. $\widehat{\beta_{42}}$ = -0.42 means keeping other factors constant, engaging in moderate physical activity decreases log odds of obesity by 0.42.
8. $\widehat{\beta_{43}}$ = -1.11 means keeping other factors constant, engaging in high physical activit**y** decreases log odds of obesity by 1.11 which is the most among the three categories of physical activity.
9. $\widehat{\beta_{5}}$ = 1.06 means keeping other factors constant, if an individual has a family history of obesity, the log odds of obesity increase by 1.06.
10. $\widehat{\beta_{6}}$ = -0.32 means keeping other factors constant, a high-calorie diet decreases the log odds of obesity by 0.32. This does not seem to be consistent.
11. $\widehat{\beta_{7}}$= 0.8 means keeping other factors constant, smokers have a higher log-odds of obesity by 0.80 compared to non-smokers.
12. $\widehat{\beta_{8}}$ = 0.36 means keeping other factors constant, A 1-unit increase in weight increases the log-odds of obesity by 0.36.

In our model, **weight, alcohol consumption, gender** and **physical activity** seem to be the most significant predictor for predicting the obesity status of an individual.

GOODNESS OF FIT

After fitting the logistic model in R, we get that-

Null deviance: 2901.13 on 2102 degrees of freedom

Residual deviance: 548.29 on 2091 degrees of freedom

Interpretation:

1. **Null Deviance:** The null deviance reflects the goodness of fit of a model that only includes the intercept (the null model), meaning it has no predictors. It shows the difference in deviance between the null model and the model being assessed. A lower null deviance suggests that the model accounts for a greater proportion of the variance in the response variable compared to the null model.
2. **Residual Deviance:** The residual deviance indicates the goodness of fit of the model after adding predictors. It represents the difference in deviance between the evaluated model and the saturated model (which perfectly fits the data by having a parameter for each observation). A smaller residual deviance signifies a better fit of the model to the data.

Here, **null deviance is 2901.13** and **residual deviance is 548.29**. Which suggest that the predictors included in the model explain a significant proportion of the variance in the response variable. So, the fitted model is a **good fit.**

QUALITY EVALUATION OF THE MODEL

We evaluate the quality of the logistic regression model by forming a **Confusion Matrix.**

Definition of a Confusion Matrix:

A **confusion matrix** is a performance evaluation tool used in classification models to assess how well the model distinguishes between different classes. It provides a 2x2 tabular summary of predictions by comparing actual and predicted values, offering insights into the types of errors the model makes. By breaking down predictions into categories such as **true positive rate (TPR), false positive rate (FPR),** the confusion matrix helps in calculating key evaluation metrics like accuracy, precision etc. This detailed breakdown enables a deeper understanding of model performance beyond a single accuracy measure.

Choice of threshold:

Then we need to chose an optimum threshold p*. The value of p* is so chosen that for p_i=p* the True Positive Rate (TPR) is high and the False Positive Rate (FPR) is low. We take p* for which **TPR*(1-FPR) is maximum.**

Where, TPR = P [predicted response = 1 | observed response = 1]

FPR = P [predicted response = 1 | observed response = 0]

Here we get our optimal threshold or cutoff point as p* = 0.5087012

= 0.5 (approximately)

By using the optimum threshold value of 0.5, we form a confusion matrix. We use the confusionMatrix( ) function from the “caret” package in R. The confusion matrix is shown in the next page.

Confusion Matrix:

Observed response / Predicted response	Non-Obese (0)	Obese (1)
Non-Obese (0)	**1079**	**59**
Obese (1)	**59**	**906**

From this matrix we calculate the TPR and the FPR.

TPR = P [predicted response = 1 | observed response = 1] = 0.9481547

Which means the model correctly identifies almost 94.8% of actual obese individuals.

FPR = P [predicted response = 1 | observed response = 0] = 0.0611399

Which means almost 5% of non-obese individuals are classified as obese.

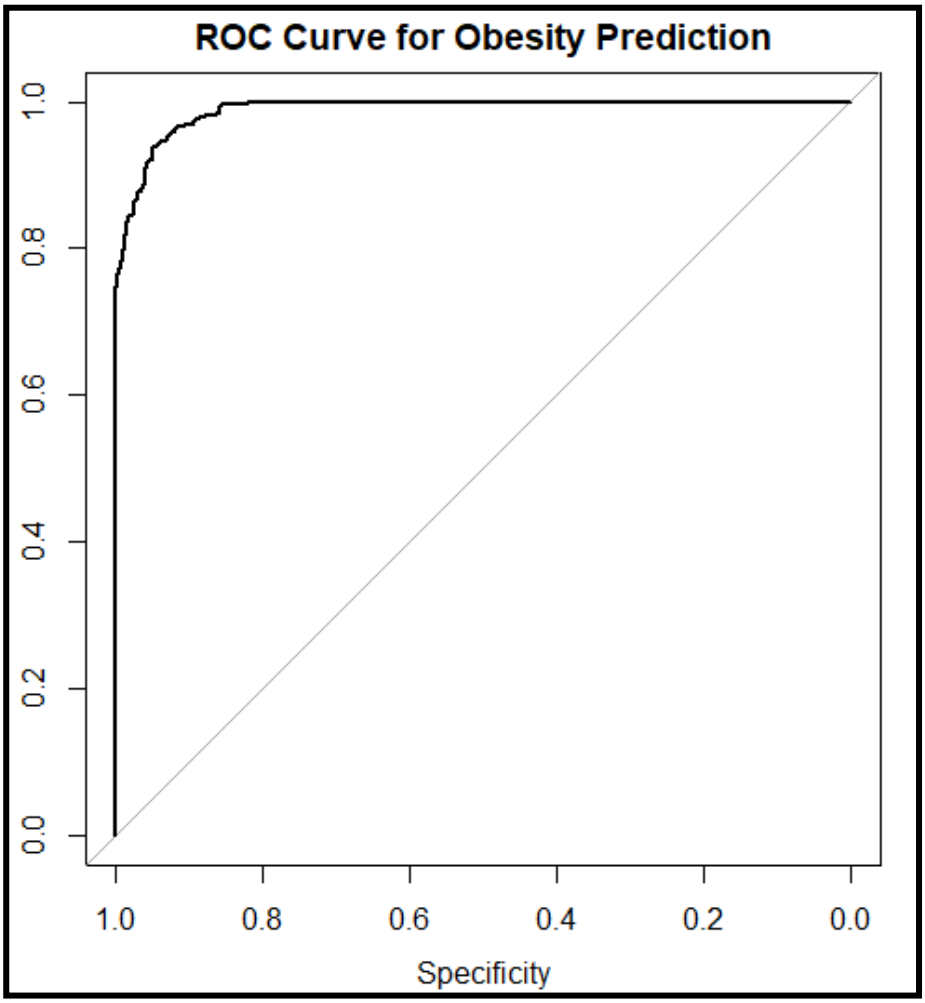
Comment:

Here we have sufficiently low FPR, so we can conclude the logistic regression model correctly detects obesity and non-obesity in most of the cases.

RECEIVER OPERATING CHARACTERISTIC (ROC) CURVE

The Receiver Operating Characteristic (ROC) curve is a graphical method for measuring the performance of a binary classification model. ROC curve is used to plot True Positive Rate (TPR) vs False Positive Rate (FPR) at different decision thresholds. The ROC curve offers information on how well the model can separate positive and negative classes at different threshold values.

The TPR is plotted on the y-axis, while the FPR is plotted on the x-axis. An ROC curve for a model that is closer to the top-left of the plot signifies high sensitivity and low false positive rate, which indicates good model performance. We obtain the ROC curve in R which is shown below.



Comment:

The ROC curve **hugs the top-left corner** of the plot which indicates the logistic model has high discriminative ability. Which means this is a good model to distinguish between obese and non-obese individuals.

If the curve were closer to the diagonal line, it would indicate poor performance, as the model would be no better than random guessing.

We also get the area under the ROC curve or the **AUC** as 0.9892. The closer the AUC is to 1, better the model.

Again, AUC = 0.9892 means **98.92% chance** that this model will rank a randomly chosen obese individual higher than a randomly chosen non-obese individual in terms of predicted probability.

CONCLUSION

In this research, we did a thorough examination of obesity to analyse factors that affect obesity and compare the performance of various statistical tools. First, we used stacked bar diagrams to map out the spread of categorical variables, providing an easy picture of demographic distribution and the prevalence of obesity in various groups. Detection of outliers was done through boxplots, from which we eliminated influential observations to guarantee the stability of our models and reduce their influence on the outcome.
To analyse associations between categorical predictors and the response variable, we computed Yule's Q, giving a measure of association for dichotomous variables. This aided in identifying the most important factors that may be related to obesity. From Q and Chi-square test, we discovered that physical activity, alcohol consumption, and family history were significant factors influencing obesity. Further, a logistic regression model was constructed to estimate the probability of obesity using the available predictors. The logistic model identified weight, gender, and physical activity as the strongest predictors of obesity. The performance of the model was measured using a confusion matrix, providing information about its accuracy. In addition, the receiver operating characteristic (ROC) curve was graphed to measure the discriminative power of the model. From the results, it was shown that the logistic model accurately portrays the relation between the predictors and obesity.

In the end, the research presents a holistic perspective on obesity, which can be useful in creating focused interventions for obesity prevention and treatment in future.

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